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Secondhand Car Recommendation for Consumers in Bangladesh Using Machine Learning

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SECONDHAND CAR RECOMMENDATION FOR CONSUMERS IN BANGLADESH USING MACHINE LEARNING

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THESIS SUBMITTED IN FULFILMENT FOR THE DEGREE OF
B. SC. IN COMPUTER SCIENCE AND ENGINEERING

DEPARTMRNT OF COMPUTER SCIENCE AND ENGINEERING (CSE)
INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG (IIUC)
CHITTAGONG, BANGLADESH

2024

DECLARATION

I/We hereby declare that the work in this thesis is our own except for quotations and summaries which have been duly acknowledged.

25 November 2024

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SUPERVISOR'S DECLARATION

I hereby declare that we have read this thesis and in my/our opinion this thesis is sufficient in term of scope and quality for the award of the degree of B. Sc. in Computer Science & Engineering

25 November 2024

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ABSTRACT

Taste, preferences, choice, and mostly financial strength are the main factors when it is a matter of hobby or it can be also called outside interest. Most of the people of our country Bangladesh, especially people of 18-50 age have so much fascination with the car. It's not an easy thing to buy a new car because of its higher value and cost. That's the reason, the maximum people of in our country nearly 80% of car users in Bangladesh buy a secondhand car or used car according to their needs. At the time of buying a used car, they have to notice so many factors to find a good one according to their budget and other preferences. For this reason, they need a good recommendation according to their preferences to find a worthy car. To solve this problem, we aim to build a recommendation system with two datasets, Bikroy1 and Bikroy2, where recommendations will be generated based on user preferences. Here we will be using traditional algorithms of machine learning like Decision Tree, Random Forest, TF-IDF, and KNN and the Learning-to-Rank type machine learning algorithm LambdaRank method to build a used car recommendation system. We aim to achieve an accuracy of more than 98% in Decision Tree, Random Forest, and KNN algorithms and an NDCG score of more than 0.98 for both datasets Bikroy1 and Bikroy2. Results obtained from the models of machine learning which is used in this research will be compared according to both datasets and analyzed to find out the better-performing dataset and model. We believe this research provides the insights and information that are needed to build a good used car recommendation system for Bangladeshi consumers.

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LIST OF ABBREVIATIONS

KNN	K-Nearest Neighbors
SVM	Support Vector Machine
DT	Decision Tree
RF	Random Forest
MAP	Mean Average Precision
LTR	Learn To Rank
NDCG@K	Normalized Discounted Cumulative Gain at K
TF-IDF	Term Frequency-Inverse Document Frequency

1 Introduction

CHAPTER I

INTRODUCTION

1.1 RESEARCH BACKGROUND

In Bangladesh, most of the families are middle class. A middle-class family finds it nearly impossible to purchase a car in Bangladesh due to the increased cost of automobiles. According to Budget of year 2024-25, buying imported cars has become more expensive than before. Due to not sufficient income,middle class cannot afford a new car.For this reason,they rely on secondhand car and because of that,second hand car is mostly hyped in Bangladesh.Second-hand car refers to a vehicle that has been alreay used by one customers and again sold to other customer.Comparing with other countries, the car market is not big.Few brands such as : Toyota,Suzuki,Honda are well known in Bangladesh.[23]. People choose all of these company as a higher priority when they wants to buy a car. But, it is not as simple as we think to purchase a used car. Purchasers confront different issues while buying a second-hand car. For buyers, they are to keep an eye on the car's model, brand, fuel type, mileage, manufacturing year, and other factors[8]. It is quite challenging for buyers to make a better choice to buy a used car while considering every constraint such as brand value, mileage, budget, and so forth. With hundreds of options available consumers can become overwhelmed with the number of choices. A personalized recommendation system could help narrow down relevant cars based on preferences[8]. Cause the features and the other preferences of user are very much important for choose an ideal car. That’s why to satisfy buyers all of their conditions and requirements a recommendation system based on Machine learning algorithms help them to take a faster decision. Also another reason of this research is there is no well accurate recommendation system in this used car E-commerce field. So this field is a proper sector for us to do a futuristic research and hope of get success highly.

1.2 PROBLEM STATEMENT

In the modern digital era , 'choice-overloaded" is a normal things. Now days there are lots of options in every sectors. Specially when it is a matter of buying a used car , a person has a lot of resources and sectors to purchase a car. But the main challenge is to interact vast amount of attributes and find a perfect match for his own. In these sectors traditional search and filter algorithm often fails to capture the unique preferences of individual user.

Content based recommendation works to tackle the issue by normalizing all attributes for used car buyers. Without building a recommendation system several kinds of problems occurs here those are given below –

1. Difficulty in Personalizing Recommendations : Different kinds of needs but integrating their together is tough.
2. Inconsistent Quality of Listings: Used car market can be varied and its data can be biased.
3. Multiple and Conflicting Buyer Preferences : Buyers preference is conflicting to the existing data's and cannot take decision worth or not.
4. Trust and Engagement : Real data reflect or real engagements reflects of buyers and sellers.
5. Cold Start Problem
6. Less open resources

To resolve all of these problems a content based recommendation system is need to handle all over things and much advanced work.

1.3 MOTIVATION

This research is motivated by the growing need for user-centered services in the E-commerce and for auto-mobiles shop. The activity of purchasing a used car is quite complex as it involves both emotions and inter rational considerations. Straightforward buying decisions, purchasing a car cannot justified the variables such as cost, brand, technical requirements, and individual consumer needs. This complexity often results irrelevant product to the buyers which doesn't fulfill their requirements. Recommendation systems have benefitted industries such as E-shop, Entertainment and E-travels & Hospitality by providing targeted suggestions to the end user. But, the E-commerce car industry and auto mobiles industruy does not benefit from these developments and in many cases users are forced to look for cars using inefficient methods that do not take into consideration their specific requirements. This study fills that gap by proposing the use of machine learning algorithms in building a recommendation system that would help

users find the appropriate cars. The major reason is the ability to emphasize such simple yet powerful techniques as Decision Tree, Random Forest, Learn To Rank, TF-IDF & Cosine-similarity, and KNN. We will try to make some effective model based on those algorithm and draw out the comparative result. More widely, such approaches have focused on textual and categorical data and can be repurposed to interrogate other data forms including car features, consumer demand and the car modeling cycle. Basically all those method can handle complex data and figure out a effective outcomes. Another motivation is to build a recommendation system that combines clarity, accuracy, and personalization can transform this experience, building trust and engagement between buyers and platforms also terminate the technical hassle or complexity. In the end, the aim is not only the greater effectiveness of E-commerce car systems but to also encourage the use of comparable ideas in other areas. This research shows that in circumstances with different decision-making criteria which often subjective, technology based solutions can bring positive changes, enhance the overall experience of the customers and set the new trends within the E-commerce.

1.4 OBJECTIVE OF RESEARCH

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The research aims to address the issue of used car recommendations by achieving the following objectives:

1. To extract important features from datasets for effective recommendations.
2. To make reasonable used car recommendations based on user preferences using machine learning models effectively.
3. To ensure good precision of recommendation and ensure they align closely with user interests.
4. To ensure that recommendation algorithm can handle large datasets.

2 Literature Review

CHAPTER II

LITERATURE REVIEW

2.1 INTRODUCTION

Buying a new car can be extremely gratifying when everything done properly. Unfortunately, the buying process can be little bit difficult to navigate for newbies[1]. There are few things which can be call variables should be kept in mind before buying second-hand car. Through the integration of crucial variables like the vehicle's age, brand, model, mileage, and price with the buyer's individual preferences like features, budget, and brand, a system must be offered more customization and effective enviourment. To bring these difficult work less many authors build many kinds of system. Car price prediction are famous among them . Besides Prediction and classification systems using attributes are also remarkable in research world. But recommendation system is much familiar in outsiders sector of vehicle. Netflix [2], Youtube[3] these kinds of platform use these system for there work.

2.2 SCOPE OF RESEARCH

There are several research gaps in the existing and previous papers. Few studies are conducted in different cultures and market contexts. Certain researches that concentrate solely on used car price prediction are directed towards Bangladeshi secondhand markets. Content-based filtering remains underutilized in this context, with limited integration of diverse attributes such as accident history, registration status, and subjective preferences There were probably no studies conducted on secondhand car recommendation systems in Bangladeshi used car market contexts with the combination of machine learning.

2.3 LITERATURE REVIEW

This research focuses on the development of new techniques for recommend used car based on the user preferences. But in previous all of our reviewed literature was maximum about predictive or regression model. Here are some details about those literatures which reviewed for build these system and generate ideas. KNN, SVM, XGBOOST ,ResNet, DXL these all previously worked for price prediction sectors. On content based filtering much review paper

showed many comparison between collaboration and collaborative where content based is the better proved[14]. Recommendation system previously used for movie recommendation in Netflix[2] and also in mostly used Youtube[3].

2.3.1 General

Peoples wants and people suggestion is a main things for build up a recommendation system. It is a important factor in day to day life. In the world of used car it is more important and noticeable things . With the help of KNN, SVM a model already build for price prediction.[4] They compared the performance of KNN and SVM to see their work and they find 83 percent for KNN and 80 percent for SVM[4]. In that price prediction sector XGBOOST also already showed its performance as a prediction model[5]. Some Chinese authors worked for that and they increased the accuracy of existing XGBOOST model and increased its performance 0.488 percent[6]. The accuracy of that model is 0.548 percent [6]. But rather than XGBOOST, ResNet and DXL performs well in price prediction one was 63 percent and other was 75 percent[11]. Recently some students from Brac build a price prediction model where XGBOOST was 90 percent and random forest showing 93 percent accuracy[15].

Two widely used recommendation systems—content-based and context-based recommendation systems are also used for recommendations[7]. Content based works over collaborative approach also showed in music recommendation in a million music's dataset[14]. Content based recommendation will helpful for a web creator to suggest person to person various dynamic window[13]. It was worked with peoples interaction on world web and these works as a dataset for them. With that they create recommendation system which is based on contest based system[7].

Every time fuzzy logic don't work appropriately with the model and integrations[9].But some fuzzy logic principles have been applied to improve recommendation accuracy, to capture the inherent uncertainty and imprecision in user preferences[8].Though in some sectors fuzzy logic work as main classification master for find out pattern related to needs[12]. Combination of Fuzzy Logic and the KNN algorithm has been applied in the recommendation system to obtain more nuanced and accurate recommendations. Though it's a recommendations of nutrition for HIV sufferers [10].

2.3.2 Specific

The development of second hand car exhortation system has been become highly focused research work due to growing demand for personifying solutions in the secondhand self propelled market. Nevertheless, existing studies frequently address determined aspects, leaving gaps in fully enhanced and context-specific recommendation frameworks. The specific work in these sectors are :

KNN , SVM based model gets higher accuracy in these sectors which are 80 percent and 83 percent[4]. XGBOOST also working for it with a good accuracy[5]. From Chinese publication the XGBOOST performance is increased and it was 55 percent almost[6]. ResNet and DXL performs more well then XGBOOST[11]. But recently a work publish from which we saw Random forest and XGBOOST are the real winner in these sector[15].Though XGBOOST is a advanced technology for these work.

Review of Content Based and others filtering

Content based works over Collaboration filtering because it has some cold start problem[14].It was examined with millions of music dataset[14].Content based also helpful for various tastebuds dynamic window[13]. Similarly it is working for people integration in world wide web[7].

Fuzzy Logic recommendation

Fuzzy filtering or fuzzy logic can be working as a algorithm in some sectors modifying its work[12].Besides combining fuzzy and KNN will represent a good accuracy in model performances[10]. Through every time fuzzy logic don't work properly with every model and integration[9].

2.4 SUMMARY

Machine learning based prediction model are available in research world with the concept of price prediction. Researchers were try with various kinds of algorithms to predict the result and obtain a good result from that. From study in the sector of price prediction the XGBOOST and random forest model is the winner.In recommendation sector we get movie recommendation system several times but they aren't up to mark as a recommendation model. In Netflix recommendation sector it makes a good impact till now. Besides integration ML with fuzzy logics represent some good results and sometimes inappropriate results in accuracy. In the recommendations sector content based is talks over collaboration because of some problems. Thats all about the literature study covered as a background of these work.

CHAPTER III

METHODOLOGY

3.1 INTRODUCTION

In recent years in Bangladesh used car exchanging rate is increasing day by day. Thats why when a person interested to buy a used car he have to face some problems. Here these system aims to provide a good recommendation to buyer which is helpful for a buyer to get a good and trusted suggestion .These model works with two datasets. Sources are bikroy . By integrating some machine learning techniques and advanced ranking algorithms here these system preprocessed data firstly and then the model do its own work to recommend a good result. Here these system works through content based filtering to suggest releavent data to the user .

These system works with DF,RF model and also with KNN as supervised model. A unsupervised model also present here which is TF-IFD and another algorithm is LamdaRank which is a ranking algorithm. These algorithm works on Bikroy1[21] and Bikroy2[22]

3.2 DATA ACQUISITION

Here in these used car recommendation system the one dataset was collected from github which is used in a paper previously . But the another one was acquired through web scraping from the online marketplace Bikroy.com, a popular platform for buying and selling vehicles in Bangladesh. It provides diverse data about a car. Here we prioritized HTML scraping technique which is helpful for us to easily extract data. Some other techniques of scraping like BeautifulSoup and Selenium which gave us some bad experience and also couldnot provide us proper data according to the requirement. We also try to extract data from Bhalogari.com. But the port of their datasets was private . That's why in that site we experienced error with every scraping techniques. In first dataset there are 2137 data with 13 attributes. Simillarly in second

dataset contains 1143 data with 13 attributes. The selected attributes which are working for these system those are –

Table 3.1 : Attribute Details

Attributes	Illustration
car_name	Name of car
brand	Brand of car
car_model	Model of car
model_year	Manufacturing year
Transmission	Automatic or munual
body_type	Body types of car
fuel_type	Fuel category of car
engine_capacity	Engine capacity in cc
travels	Kilometer travels by car
price	Asking Price of it
accident	Ever faced any accident
registration	Papers validation

3.3 DATA PRE-PROCESSING

Before data analysis, they must go through data preparation which consists of organization, transformation and data clarification. This work should be done before data evaluation. It includes removing invalid values, missing values, as well as reformation data, scaling variables, and creation of new variable. Here in these system CCA methods is used for Handling missing values and imputation techniques. According to that method if the loss is less than 5% then it removes those data and in imputer it does mean operation. Also here we remove null values via remove row if its duplicated data.

Unique Value

Table 3.2 : Unique Value(Bikroy1)

Attributes	Unique Value
car_name	1108
Brand	26
car_model	123
model_year	35
Transmission	2
body_type	10
fuel_type	6
engine_capacity	50
travels	638
Price	506
Accident	2
registration	6

Table 3.3 Unique Value(Bikroy2)

Attributes	Unique Value
car_name	1108
brand	26
car_model	123
model_year	35
Transmission	2
body_type	10
fuel_type	6
engine_capacity	50
travels	638
Price	506
Accident	2
registration	6

Null Value Before Cleaning :

Table 3.4: withnull(Bikroy1)

Attributes	Null
car_name	0
Brand	0
car_model	0
model_year	0
Transmission	0
body_type	0
fuel_type	0
engine_capacity	0
travels	0
Price	0
Accident	0
Registration	0

Table 3.5 : withnull(Bikroy2)

Attributes	Null
car_name	0
Brand	0
car_model	0
model_year	0
Transmission	0
body_type	0
fuel_type	0
engine_capacity	0
travels	2
price	14
Accident	0
registration	0

Null Value After Cleaning :

Table 3.6 : without null(Bikroy2)

Attributes	Null
car_name	0
Brand	0
car_model	0
model_year	0
Transmission	0
body_type	0
fuel_type	0
engine_capacity	0
travels	0
price	0
Accident	0
registration	0

3.4 METHODOLOGY

Here the system is based on content based filtering . For make good movement of content based filtering some machine learning models are used in background of it . Those models are Decision Tree , Random Forest , Learn to Rank , Term Frequency-Inverse Document Frequency(TF-IDF) , Naïve Bayes . Here content based filtering with these models working

with the attributes such as brand, car_name, car_model, model_year, transmission, body_type, fuel_type, engine_capacity, travels, price, accident, registration. **Bikroy1** & **Bikroy2** named two datasets working here.

3.4.1 USED MODELS

DECISION TREE

Decision tree algorithms are a type of supervised learning method used for classification problems. It can work with both numerical and categorical data. It creates tree of decisions and allowing prediction with based on several input features[16]. Here in these system decision tree algorithm starts at the root node and traverses through the tree by making a decision based on the input feature values, until it reaches a leaf node. The value at the leaf node represents the predicted output value[16]. Here these algorithm calculate information gain or gini for each features. The highest information gain of a feature then again split into childs. These process continues until a stopping criterion meets. In calculation pi is the probability of an instance being classified into a particular class and Di is the subset of D after splitting by an attribute[17].

$$\text{Gini} = 1 - \sum \ln(\text{pi})^2 \qquad \text{---- (3.1)}$$

$$\text{Entropy} = -\sum (\text{pi}) \log_2(pi) \qquad \text{---- (3.2)}$$

$$\text{Information Gain} = \text{Entropy}_{\text{ parent}} - \sum \left(\frac{Di}{D} * \text{Entropy(Di)}\right) \qquad \text{----- (3.3)}$$

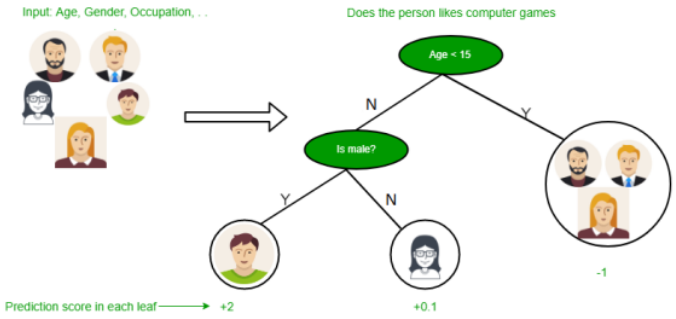


Figure 3.1 : Decision Tree

3 RANDOM FOREST

Random Forest classification is a form of ensemble learning method that makes predictions by using several decision trees. It is a change to the Random Forest algorithm, which is often used to sort things into different groups[15]. It is a ensemble learning create multiple subset form training data as a decision tree. Each tree has a leaf value or answer. Here every tree calculate a classification of data. Lastly the mostly counted or most popular answers from all of the trees will be consider as output. Here bagging makes all results different from each other. The voting here will run through the user preferences. Most decision trees output are consider as result of model.

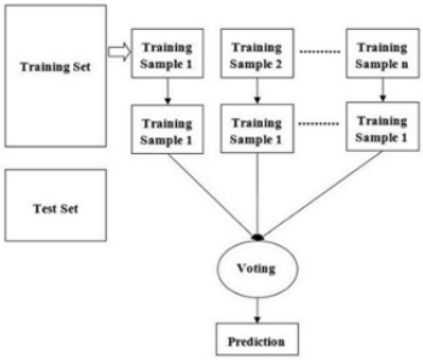


Figure 3.2 : Decision Tree

TF-IDF VECTORIZATION WITH COSINE SIMILARITY

It recommend most recommend options similar to user preference. It is a good classification which measures similarity depends on vector[18]. It takes user preferences as a query and convert it to a vector. Then compute similarities using cosine similarity[18]. Before it the model measures term frequency and Inverse document frequency for datasets and compute there vectors.

$$\text{TF}(t, \text{text}) = \frac{\text{Number of times item appear in text}}{\text{total numbers of item in text}} \quad \text{--- (3.4)}$$

$$\text{IDF}(t,\text{corpus}) = \log\left(\frac{\text{Total number of text in corpus}}{\text{number of texts containing term t}}\right) \quad \text{---- (3.5)}$$

$$\text{TF-IDF}(t,\text{text},\text{corpus}) = \text{TF}(t,\text{text}) \times \text{IDF}(t,\text{corpus}) \quad \text{---- (3.6)}$$

$$\text{Cosine Similarity}(A,B) = \frac{A.B}{\|A\| \times \|B\|} \quad \text{---- (3.7)}$$

K-NEAREST NEIGHBORS

It is a straightforward yet effective algorithm for content based recommendation system. It totally based on similarity score which is searching by a user. It firstly measure Euclidean distance between user preference and training data. Then calculate similarity of features. If similarity is greater then threshold then it will give us that data as recommendation.

$$\text{Euclidean Distance} = \sqrt{((X - B_1)^2 + (Y - B_2)^2)} \ [(X,Y) \text{ test features}] \quad \text{---- (3.8)}$$

$$\text{Similarity} = \text{Matching features/Total features} \quad \text{----- (3.9)}$$

$$\text{Good recommendation} = \text{threshold} < \text{Similarity} \quad \text{------(3.10)}$$

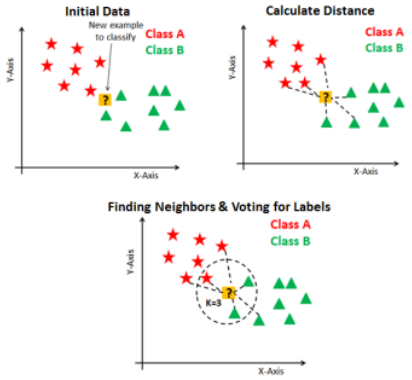


Figure 3.3: KNN algo

LEARN TO RANK (LAMDA RANK)

It is a class of supervised learning algorithms it sorting lists in terms of releavence of a query. It predicts a score from given input. Here input data are classified into different classes which means rank. It measure ranking score for all classes. Then user preferences here also divert into ranking scores[19]. It compare relavence score pairwise and rank them in different class. For each pair of items its calculate gradient which called lamda.

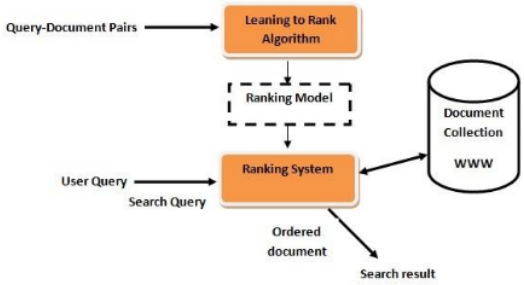


Figure 3.4 : LamdaRank

3.4.2 WORKFLOW OF MODELS

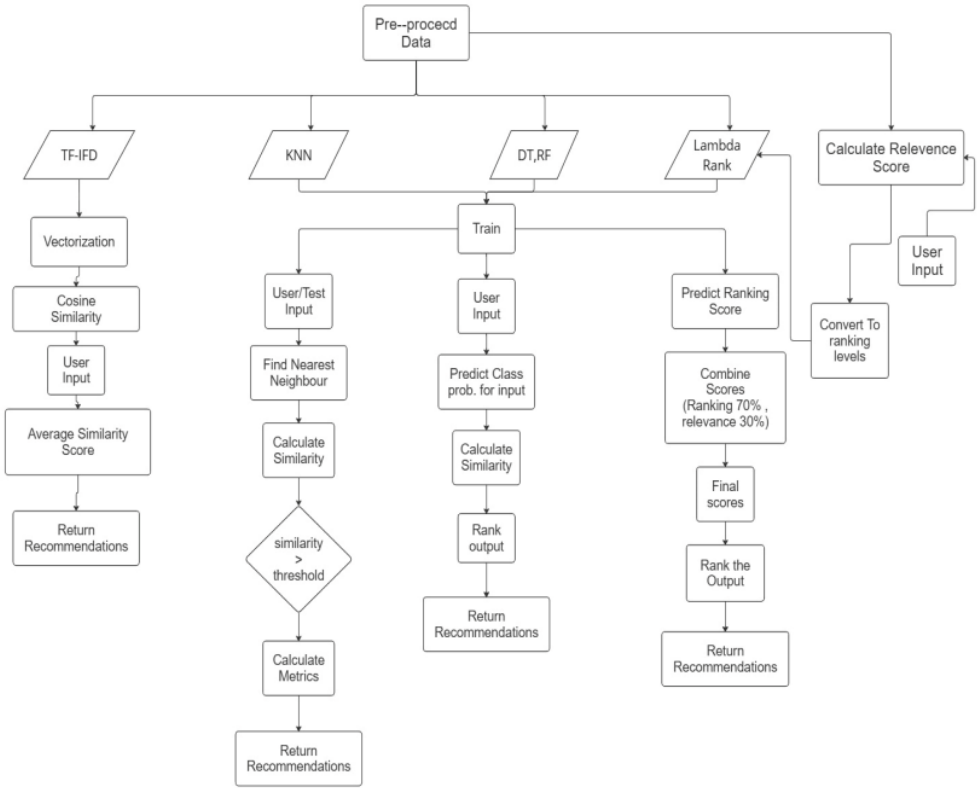


Figure 3.5 : Workflow of all model

CHAPTER IV

RESULTS AND DISCUSSION

4.1 INTRODUCTION

This study investigates a secondhand car recommendation system for consumers in Bangladesh utilizing Machine Learning and Fuzzy Logic. Modeling a secondhand car recommendation system in Bangladesh presents a significant challenge, as Bangladesh is a developing country with a substantial middle-class population. The research employs two datasets, **Bikroy1**[21] and **Bikroy2**[22], which contain secondhand car features.

4.2 ANALYSIS

This study will compare the performance of two datasets, **Bikroy1** and **Bikroy2**, utilizing traditional ML methods such as Decision Tree and Random Forest, as well as Learning-to-Rank (LTR) employing the LambdaRank technique, KNN, TF-IDF & Cosine-similarity. The experiment aims to provide insights into the efficacy of these two datasets in generating car recommendations based on user preferences, evaluating their performance using various machine learning methods with respect to accuracy and performance metrics. Both datasets were randomly divided into training and test sets, with 80% for training and 20% for testing.

4.2.1 ANALYSIS OF RESULT FOR DT & RF

In this model, we use DT and RF model together to recommend secondhand car for both datasets, Bikroy1 and Bikroy2. For dataset Bikroy1, the DT and RF model the accuracy and others evaluation metrics such as precision, recall and f1-score is given by figures.

Table 4.1 Accuracy for DT & RF

Dataset	Decision Tree	Random Forest
Bikroy1	98.83%	98.83%
Bikroy2	100%	99.563%

Table 4.1 presents the accuracy of our random forest and decision tree model that correspond to datasets Bikroy1 and Bikroy2.

PERFORMANCE ON DIFFERENT ACCURACY

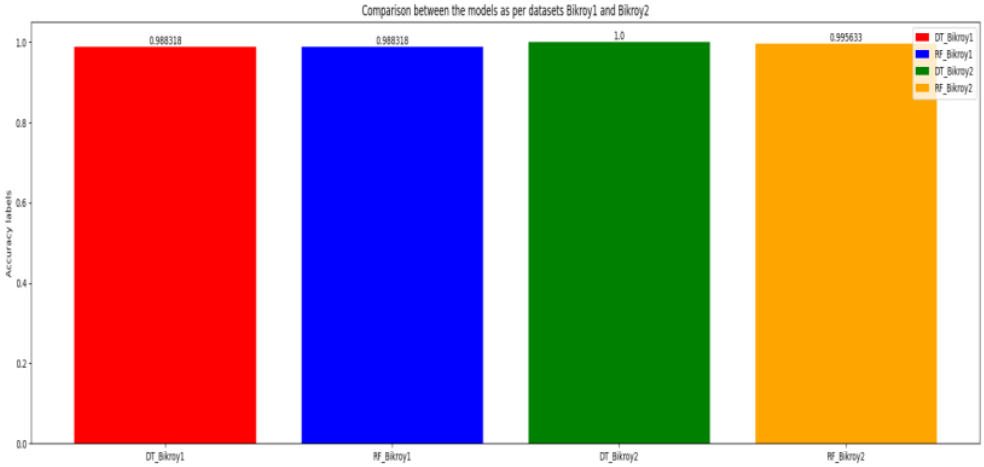


Figure 4.1: (Result Comparison)

Above Figure 4.1 describes accuracy labels in DT and RF algorithms as per both datasets, Bikroy1 and Bikroy2. These describes in DT and RF algorithm achieve 100% and 99.563% accuracy when it works with Bikroy2 dataset and performs better for Bikroy2 dataset than Bikroy1 dataset.

CLASSIFICATION REPORT ON DATASETS

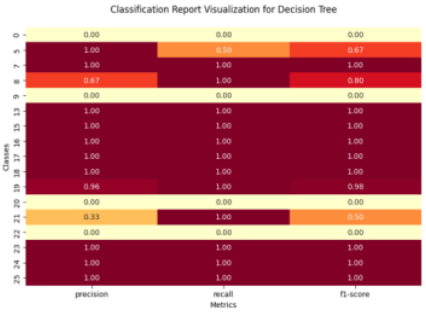


Fig 4.2 DT classification using Bikroy1

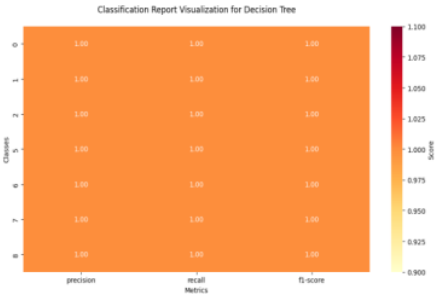


Fig 4.3 DT classification using Bikroy2

Above figures Fig 4.2 and Fig 4.3 demonstrates the classification reports for Bikroy1 dataset and Bikroy2 dataset for DT Model. Here, we use “classification_report” method from “sklearn.metrics” for better performance and to obtain more precise values of precision, recall

and f1-score. Here, we can observe the evident differences between classification reports of DT for the datasets Bikroy1 and Bikroy2 in these figures.

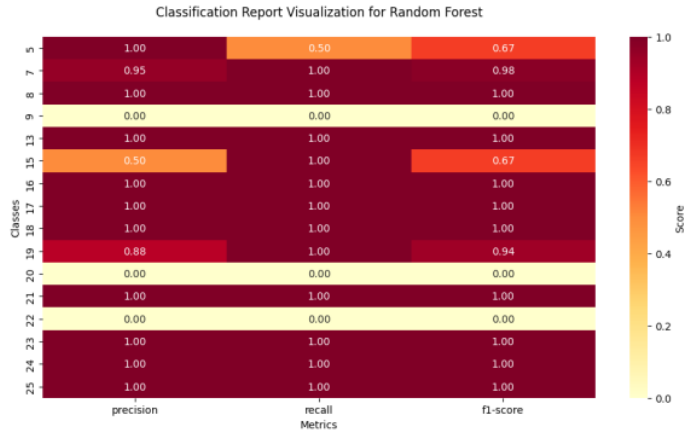


Fig 4.4 RF classification using Bikroy1

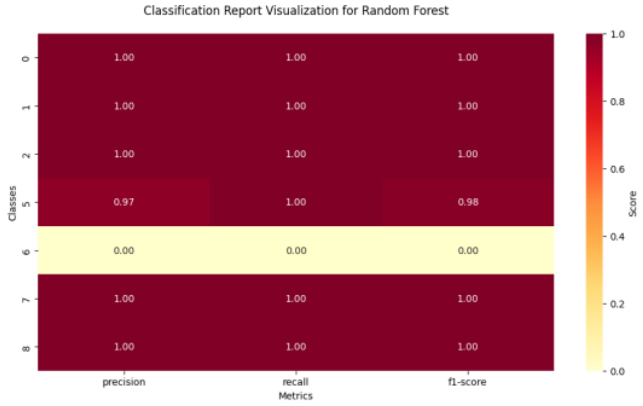


Fig 4.5 RF classification using Bikroy2

Above figures Fig 4.4 and Fig 4.5 demonstrates the classification reports for Bikroy1 dataset and Bikroy2 dataset for RF model. Here, we use “classification_report” method from “sklearn.metrics” for better performance and to obtain more precise values of precision, recall and f1-score. Here, we can observe the evident differences between classification reports of RF for the datasets Bikroy1 and Bikroy2 in these figures.

RECOMMENDATIONS FOR DIFFERENT DATASETS

Here we will illustrate the secondhand car recommendations for both datasets Bikroy1 and Bikroy2 and analyze how most similar secondhand cars are being selected for

recommendations in an effective manner. In this study we analyze how recommendations are generated using datasets and user preferences obtained from the user end.
User preference is a set of personal data of the unique choices that an individual can make within a software application, websites or digital platform.

Table 4.2: Example user preferences

Attribute	Value
Model Year	2020
Transmission	Automatic
Body Type	SUV
Fuel Type	CNG
Engine Capacity	1500 cc
Travels	30,000 km
Price	45,000
Accident History	No
Registration Year	2024
Brand	Toyota

Table 4.2 which is depicted in the above demonstrates an example user preferences and it is used to generate preowned car recommendations.

Table 4.3: Similarity score for Bikroy1 datasets

dataset_index	similarity_score_dt	similarity_score_rf	similarity_score_combined
2134	100.0	94.00	97.0000
0	100.0	94.00	97.0000
1	100.0	94.00	97.0000
2133	100.0	94.00	97.0000
2132	100.0	94.00	97.0000
...	...	...	...
178	0.0	0.03	0.0075
1983	0.0	0.03	0.0075
2077	0.0	0.02	0.0050
847	0.0	0.02	0.0050
812	0.0	0.00	0.0000

Above Table 4.3 reports the similarity score as per Decision Tree, Random Forest and combined and demonstrates the most similar secondhand car details from Bikroy1 dataset which was obtained by calculating the similarity scores using user preferences and Bikroy1 dataset.

Table 4.4: Top five recommendations for Bikroy1 dataset

dataset_index	car_name	brand	car_age	price	travels
2134	Toyota Alphard 2006	Toyota	16	3000000	206693
0	Toyota Fielder 2011	Toyota	8	1200000	45852
1	Toyota Noah 2005	Toyota	14	1200000	219630
2133	Toyota Alphard 2006	Toyota	16	3000000	206693
2132	Toyota Alphard 2006	Toyota	16	3000000	206693

Table 4.4 illustrated in the above demonstrates the top five used car recommendation based on their similarity scores and delineates the fact that most similar cars were selected for recommendation according to their similarity scores. Here, the brand of the recommended cars are seemed to be same and only five features of the car are being shown for the simplicity.

Table 4.5: Similarity score for Bikroy2 dataset

Dataset Index	Similarity Score (DT)	Similarity Score (RF)	Combined Similarity Score
1142	100.0	99.0	99.5
1141	100.0	99.0	99.5
1	100.0	99.0	99.5
2	100.0	99.0	99.5
3	100.0	99.0	99.5
...	...	...	...
620	0.0	0.0	0.0
577	0.0	0.0	0.0
353	0.0	0.0	0.0
267	0.0	0.0	0.0
311	0.0	0.0	0.0

Above Table 4.5 reports the similarity score as per Decision Tree, Random Forest and combined and demonstrates the most similar secondhand car details from Bikroy2 dataset which was obtained by calculating the similarity scores using user preferences and Bikroy2 dataset.

Table 4.6: Top five recommendations for Bikroy2 dataset

dataset_index	Car Name	Brand	Car Age	Price	Travels
1142	Toyota Fielder Wxb Mica Blue 2019	Toyota	5	2,880,417	70,759
1141	Toyota Fielder Wxb Mica Blue 2019	Toyota	5	2,880,417	70,759
1	Toyota Hiace 2005	Toyota	19	2,000,200	32,000
2	Toyota Axio 2009	Toyota	15	2,245,550	25,500
3	Toyota Corsa 1999	Toyota	25	1,120,000	90,000

Table 4.6 illustrated in the above demonstrates the top five used car recommendation based on their similarity scores and delineates the fact that most similar cars were selected for recommendation according to their similarity

4.2.2 ANALYSIS FOR THE RESULT OF LAMBDARANK MODEL

The method known as LambdaRank is an LTR. The LTR class of supervised machine learning algorithms aims to rank a list of items based on their relevance to a question. The query we use is user preferences. We examined this model with both dataset Bikroy1 and Bikroy2. NDCG and MAP are two examples of prefix ranking metrics that LTR techniques use to optimize the ranking objective function [20].

Table 4.7: LTR metrics for dataset Bikroy1

	Train Metrics	Test Metrics
NDCG@5	1.0	0.99154
NDCG@10	0.99998	0.9867
MAP	1.0	0.4

The LTR metrics, NDCG@5, NDCG@10, and MAP, for dataset Bikroy1 are described in the above table. NDCG@5 is 1.0, NDCG@10 is 0.99998, and MAP is 1.0 for the train data. However, for test data, MAP is 0.4, NDCG@5 is 0.99154, and NDCG@10 is 0.9867. It follows that train data is clearly superior to test data for our model.

Table 4.8: LTR metrics for dataset Bikroy2

	Train Metrics	Test Metrics
NDCG@5	1.0	0.9999
NDCG@10	0.99998	0.9998
MAP	1.0	0.1

The dataset Bikroy2's LTR metrics, NDCG@5, NDCG@10, and MAP, are described in the above table. For train data, MAP is 1.0, NDCG@5 is 1.0, and NDCG@10 is 0.99998. In contrast, MAP is 0.1, NDCG@5 is 0.9999, and NDCG@10 is 0.9998 for test data. It is clear from this that our model performs better on train data than test data.

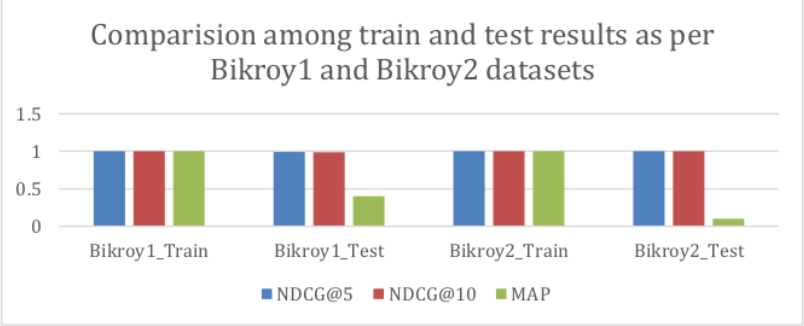


Fig 4.6: Comparison among two datasets

As seen in the above image, our model outperforms dataset Bikroy1 in terms of MAP measures on test data and dataset Bikroy2 in terms of NDCG@5 and NDCG@10 metrics. Training results for both datasets are identical.

RECOMMENDATIONS FOR DIFFERENT DATASETS

In this section, we will demonstrate the used car recommendations for the Bikroy1 and Bikroy2 datasets and examine how the most comparable used vehicles are efficiently chosen for suggestions. Our research focuses on investigating the process of generating recommendations using datasets and user preferences collected from user side.

Table 4.9: Example user preferences

Attribute	Value
Brand	Audi
Price	38,000
Travels	35,000
Car Age	2019
Engine Capacity	1,700
Accident	No

Table 4.9, which is used to provide suggestions for used cars, is shown in the above sample of user preferences.

Table 4.10: Ranking score and relevance score for Bikroy1 dataset

Data Index	Ranking Score	Relevance Score	Final Score
1480	1.000000	1.000000	1.000000
1481	0.999976	1.000000	0.999983
837	0.658178	0.996536	0.759685
834	0.652876	0.999685	0.756918
1504	0.644148	0.993073	0.748826

Table 4.10 shows the top five data for ranking scores and relevance scores for dataset Bikroy1. It delimitates the top five relevant cars using user preference and Bikroy1 dataset. Here, final score was calculated by combining the ranking score and relevance score.

Table 4.11: Top five recommendations for Bikroy1 dataset

Data Index	Car Name	Brand	Car Age	Price	Travels
1480	Nissan Sunny 1994	Nissan	30	120,000	115,000
1481	Nissan Sunny 1994	Nissan	30	120,000	115,000
837	Daihatsu Rocky 1989	Daihatsu	35	175,000	61,000
834	Kia 1994	Kia	30	125,000	3,085
1504	Chery Car Nice Condition 2006	Chery	18	230,000	72,180

In the Table 4.11, top five used car recommendations are demonstrated according to their final scores for Bikroy1 dataset. Here, it is noticeable that recommended cars are of different brands and recommended by their relevancy with the user preferences.

Table 4.12: Ranking score and relevance score for Bikroy2 dataset

Data Index	Ranking Score	Relevance Score	Final Score
38	1.000000	1.000000	1.000000
16	0.652981	0.999047	0.756801
14	0.650799	0.999047	0.755273
1101	0.650147	0.998977	0.754796
17	0.641474	0.999047	0.748746

Top five ranking and relevance scores for dataset Bikroy2 are displayed in Table 4.12. It uses the Bikroy2 dataset and user preference to define the top five relevant automobiles. Here, the ranking score and relevance score were combined to determine the final score.

Table 4.13: Top five recommendations for Bikroy2 dataset

Data Index	Car Name	Brand	Car Age	Price	Travels
38	Toyota Fielder X Octane Drive 2017	Toyota	7	80,560.0	25,600.0
16	Toyota Harrier ADVANCE 2019	Toyota	5	2,000,900.0	49,000.0
14	Toyota Harrier 2019	Toyota	5	2,000,500.0	15,000.0
1101	Toyota Fielder G 2014	Toyota	10	2,142,612.0	51,514.0
17	Toyota Harrier ADVANCE 2019	Toyota	5	2,000,900.0	49,000.0

Top five used automobiles suggestions based on final score for Bikroy2 dataset are shown in table 4.13. It is evident that are recommended here are of same brand and are chosen by how well they suit the user preferences.

4.2.3 ANALYSIS FOR THE RESULT OF KNN

Here in KNN the user preferences calculate similarity with the nearest neighbors. The similarity score is the main things here to calculate and it compare its encoded value with its neighbor value for recommendation. Here the accuracy scores for these one are given below-

Table 4.14: KNN result in two datasets

Dataset	Accuracy	Precision	Recall	F1_score
Bikroy1	0.99%	0.97%	0.83%	0.89%
Bikroy2	0.998%	0.998%	0.928%	0.961%

RECOMMENDATIONS FOR DIFFERENT DATASETS

Here are the overview of the system how the recommendation are setting up here. How these KNN model illustrate recommendations according to the user input. Here these system also calculate Euclidean distance by which it calculate the distance is showed .

Table 4.15: Example user preferences for Bikroy1

Value	Attribute
body_type	Saloon
transmission	Automatic
engine_capacity	1500cc
price	3000000
car_age	6

Table 4.15, which is used to provide suggestions for used cars, is shown in the above sample of user preferences.

Table 4.16: Top five recommendations for Bikroy1 using KNN

Index	Model	Price	Model_year	Body_type	Engine_cc	Travels	Fuel_type	Similarity
1299	Allion	3000000	2018	Saloon	1500	30000	Octane	1.00000
2031	Allion	2980000	2018	Saloon	1500	40300	Petrol	0.86667
2032	Allion	2980000	2018	Saloon	1500	40300	Octane	0.86667
2038	Allion	2970000	2018	Saloon	1500	37873	Octane	0.80000
2037	Allion	2970000	2018	Saloon	1500	37873	Petrol	0.80000

Table 4.16 showing us the top five recommendation based on Knn model which is used in these recommendation system. It measures Euclidean distance for each index. Mainly it measures similarity for each rows with respect to user preferences. Then the similarity score of a row or a data is mean of all attributes of a data. Highest similarity score will appear here in top.

Table 4.17: Example user preferences for Bikroy2

Value	Attribute
body_type	Saloon
transmission	Automatic
engine_capacity	1500cc
price	3050000
car_age	4

Here Table 4.17, which is used to provide suggestions for used cars, is shown in the above sample of user preferences.

Table 4.18: Top five recommendations for Bikroy2 using KNN

Index	Model	Price	Model_year	Body_type	Engine_cc	Travels	Fuel_type	Similarity
129	Axio	3340000	2020	Saloon	1500	48000	Octane	0.98899
128	Axio	3340000	2020	Saloon	1500	48000	CNG	0.98899
219	Allion	3450000	2020	Saloon	1500	1780	Octane	0.98481
218	Allion	3450000	2020	Saloon	1500	1780	Hybrid	0.98481
1095	Axio	2275000	2014	Hatchback	1500	82111	Petrol	0.96756

Table 4.17 showing us the top five recommendation based on Knn model which is used in these recommendation system. It measures Euclidean distance for each index. Mainly it measures similarity for each rows with respect to user preferences. Then the similarity score of a row or a data is mean of all attributes of a data. Highest similarity score will appear here in top.

Here we can see the comparison of pairwise neighbor. Here Honda and 2014 year can be matched cause these types of data present in data frame. But sometimes Honda 2014 cannot be matched cause it can be also compared with more preferences.

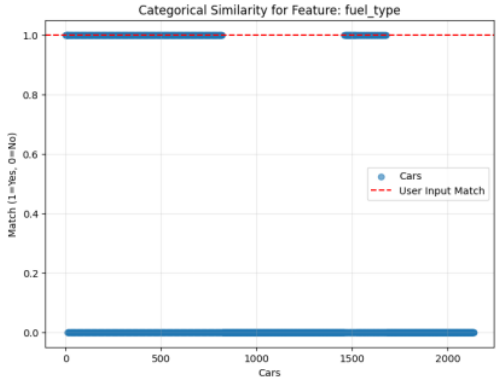


Fig 4.7 : Categorical Similarity in KNN

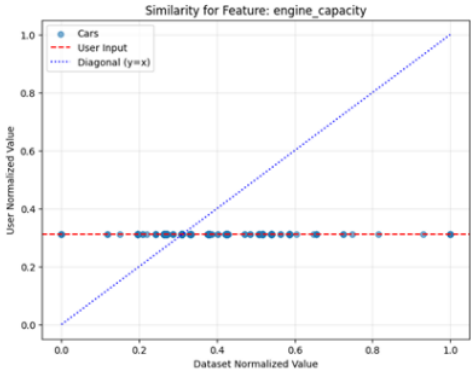


Fig 4.8 :Numerical feature Similarity in KNN

As seen in the above images in categorical data it just make encoded matches for those values. But in terms of numerical features it measures with normalized value between two sides and the point matched perfectly it will present X-Y, means that's the perfect match.

4.2.4 ANALYSIS FOR THE RESULT OF TF-IFD

Here it calculates weights to terms based on their frequency which is relative to data frame frequency across all documents and then calculate high weight , less weight. Then cosine similarity measures similarity measures angle between vectors. It's a unsupervised learning that's why here similarity scores vary from user to user.

Table 4..19 Accuracy Of TF-IFD

Preferences	Similarity Score Bikroy1	Similarity Score Bikroy2
Highest Price : 2000000	76.55%	27.09%
Maximum Age : 10		
Kilometer Travel :15000		
Minimum Engine capacity : 2500		

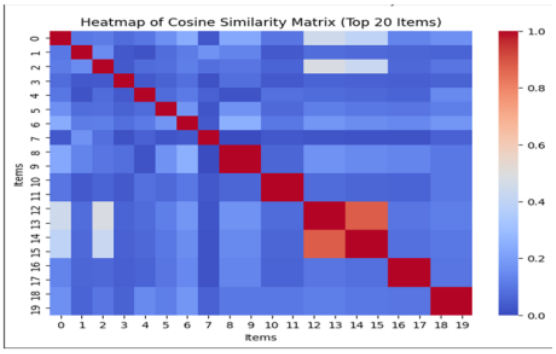


Fig 4.9 : Cosine Similarity Heatmap for Bikroy1

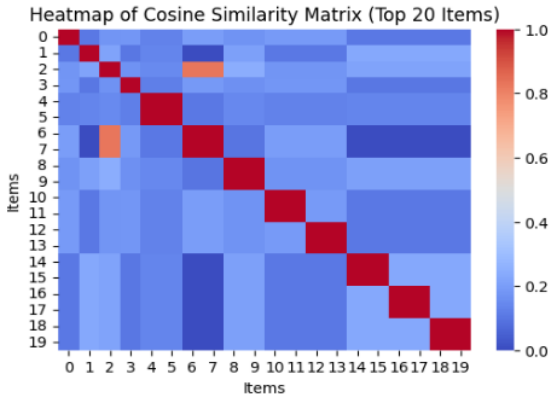


Fig 4.10: Cosine Similarity Heatmap for Bikroy2

In the above two Heatmap the more color instances means more cosine similarity means their value is highly similar , on the other hand light color denotes low similar. It present good accuracy in large dataset.

Table 4.20: Top five recommendations for Bikroy1 using KNN

ID	Car Name	Price	Engine Capacity (cc)	Age
832	Toyota Hiace GL DIESEL AMBULANCE 2015	1550000	3000	9
1887	Toyota Hiace Ambulance 2015	1390000	2000	9
1886	Toyota Hiace Ambulance 2015	1390000	2000	9
1438	Toyota Hiace AMBULANCE 2015	970000	1500	9
827	Toyota Hiace Diesel 2005	1290000	2500	19

Table 4.21: Top five recommendations for Bikroy2 using KNN

ID	Car Name	Price	Engine Capacity (cc)	Age
562	Nissan Rogue 2016	1351295.0	2500	8
782	BMW 5 Series 2016	1616458.0	3000	8
380	Mitsubishi Eclipse 2018	910277.0	2500	6
798	Mitsubishi ASX 2024	1884091.0	3500	0
414	Nissan Murano 2017	1896646.0	2500	7

In Tabe 4.17 and in Table 4.18 it shows the recommendation of cars or data based on TF-IFD and cosine similarity.

4.2.5 COMPARISON WITH PREVIOUS WORK

Table 4.22: Comparison With Previous work

Datasets	**PROMETHEE**	**DT(weighted avg)**	**RF(weighted avg)**	**KNN**
Precision for Website A Data	0.94	N/A	N/A	N/A
Precision for Website B Data	0.93	N/A	N/A	N/A
Precision for Website C Data	0.91	N/A	N/A	N/A
Precision for Bikroy1 Dataset	N/A	0.99	0.98	0.98
Precision for Bikroy2 Dataset	N/A	1.0	0.99	0.99
Average Precision	**0.93**	**0.99**	**0.98**	**0.98**
Recall for Website A Data	0.93	N/A	N/A	N/A
Recall for Website B Data	0.92	N/A	N/A	N/A
Recall for Website B Data	0.91	N/A	N/A	N/A
Recall for Bikroy1 Dataset	N/A	0.99	0.99	0.83
Recall for Bikroy2 Dataset	N/A	1.0	1.0	0.92
Average Recall	**0.92**	**0.99**	**0.99**	**0.88**

4.3 DISCUSSIONS

The analysis of both the Bikroy1[21] and Bikroy2[22] datasets underscores the notable performance improvements achieved by our proposed content based recommendation.

Here the highest accuracy obtain by the Decision tree which achieves 100% in Bikroy2 and Random forest achieves 99% . These two models recommend well with 80% train data and 20% test dataset. Similarly in learn to rank algorithm LamdaRank also done a very good job. For NDCG@5 and for NDCG@10 it obtain good accuracy in training data. Though content based recommendation system is a supervised recommendation system which always works with train data. Here KNN also achieves good accuracy which represent nearest neighbour and its accuracy is 98% for one dataset and for Bikroy2 it 99%. That means it brings up a good recommendation system. In Tf-IDF it also work well for recommendation though it is a unsupervised learning here it measures cosine similarity to find similarity . Vectorization also a factor here. Overall Random forest and KNN works better for here. It works better for categorical data in terms of recommendations.

4.4 SUMMARY

In summary, the comparison across datasets reveals that KNN, Decision Tree & Random forest these three represent good recommendation and manage all of the datasets beautifully. Here KNN and Random forest are consistent across the datasets with their model work. But here decision tree's result vary across datasets. Besides here TF-IFD with cosine similarity represent a new similarity based model which is unsupervised. The advanced thechnique of these system named LamdaRank work slightly better in train data from test data. It demonstrating its value at recommendation sectors with its advanced calculations.

CHAPTER V

CONCLUSION AND FUTURE WORKS

5.1 CONCLUSION

In summary, this research introduces a content based model, designed to enrich the boundary of recommendation sector. It is a machine learning based recommendation system which is combined several models which can be categorized into supervised learning, unsupervised learning , ranking system. These system obtain highest accuracy in supervised learning models. Though a recommendation system always a supervised technique. With these system human can take decisions on his own at the time of buying a used car. It's a content based recommendation technique which open a new way in used vehicle world. Here in Bikroy1 there are samples over 2300. Though that's not enough for a robust recommendation tasks but Bikroy1 and Bikroy2 describes well how the system work among the datasets. These system helps to people to get a personalize system to understand used car market . After all of these shortly it can be said that these work not only enhances user satisfaction but also proves its consistency in diverse dataset. It work make these system reliable for user and effective solution for used car buying sectors.

5.2 CONTRIBUTION OF THIS THESIS

These thesis makes a big contributions in the field of machine learning based recommendation system. Now buying used car is a common things across the world. Maximum time people cannot take proper decision in lack of guidelines and recommendation. So it helpful for persons to adjust better their needs with their thinking. Recently XGBOOST growing a good margins in the world of recommendation system. Here these system a LamdaRank model worked which is a advanced ranking technique with a good framework of its own. If we say specifically –

Firstly , these thesis bridges the big gap between traditional recommendation system and modern machine learning techniques. Besides it also create a bridge with ranking algorithm.

Here it ensuring a good accuracy in terms of recommendations. Because of its user friendly behaviour its easy to use and easy to understand in full scale.

Secondly, these thesis also shows the versatility of some content based recommendation model which is helpful for acquire good accuracy in recommendation world . These algorithms showing their versatility in diverse datasets.

Thirdly, it provides a detailed and advanced framework , utilizing mectrics NDCG, precision, recall which allows these system good assessment of recommendation system performances. It also evaluated potential learn to rank algorithm which is more context aware and also effective.

After all of these it can be said that these thesis contributes in the intelligent recommendation system field by performing partial, scalable and also high performing solutions.

5.3 FUTURE SCOPE

Future research will aim to push the performances of our model on large scale with large scale of dataset. Handling with much more attributes and much more categories of vehicle. Reducing model complexity will be a key focus in these sector. Integrations of user feedback make these model more effective. Deep learning integrations will reduce limitations. Cross platform recommendations also a sector to take it into a large scale system. Global adoption will be focused in future. Integration of Artificial intelligence sector also a noteable sector for future.

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